

FIFTH SEMESTER DIPLOMA EXAMINATION IN TOOL & DIE ENGINEERING

PLASTIC MOULDING TECHNOLOGY

(Maximum Marks : 100)

Time : 3 Hours

PART A

I. Answer the following questions. All questions carry two marks

- 1) Classify Plastics.
- 2) Define insert.
- 3) Define nozzle.
- 4) What is runnerless mould?
- 5) What is sputtering?

(5x2=10)

PART B

II. Answer any five of the following.

- 1) Explain the properties and uses of plastics.
- 2) What do you mean by ejection system? Explain any two.
- 3) Describe important factors of thermoplastics. Give examples.
- 4) What is a bolster? List out the various types of bolsters.
- 5) Explain injection blow moulding.
- 6) Explain two types of manifold block of hot runner unit mould with simple sketches.
- 7) Describe the process of flame and arc spraying and laser marking of decoration of plastic parts.

(5x6=30)

PART C

III. Answer one full question from each module

MODULE-I

- a) Distinguish between thermosetting materials and thermoplastic materials. 8
- b) Explain the stretch blow moulding with sketches. 7

OR

IV.

- a) Write the purpose for which following are used in plastics
1. Plasticizers 2. Fillers 3. Stabilizers 4. Resin 8
- b) With the aid of a sketch explain reaction injection moulding process. 7

V.

MODULE-II

- a) Sketch a typical mould indicating sprue bush, register ring guide pillars and guide bushes. 8
- b) Sketch and describe a typical feed system to feed four cavity mould. 7

OR

VI.

- a) With a neat sketch explain the working of an injection (screw type) moulding machine. 8
- b) What is a gate? Sketch and describe any two types of gates used in injection moulding. 7

VII.

MODULE-III

- a) Explain finger cam actuation (split in closed position) and Dog-leg cam actuation (split in closed position). 8
- b) Sketch and explain triple-daylight underfeed-stripper plate mould. 7

OR

VIII.

- a) Explain the design features of hot runner unit mould. 8
- b) What is a form pin? Draw two form pins for undercut components. 7

IX.

MODULE-IV

- a) Describe Compression moulding and Blow moulding with advantages and disadvantages. 8
- b) Explain the working principle of Transfer moulding with a neat sketch. 7

OR

X.

- a) Explain the characteristics and uses of following decoration of plastic parts 8
 - 1) vacuum metalizing 2)dipping 3)film insert marking
- b) Write short notes on 7
 - 1) sublimation printing 2)vapour polishing 3)plating